AGENTIC AI
CONTENT GENERATION SYSTEM

CREWAI • PYTHON • MULTI-AGENT WORKFLOW

PROJECT IDEA
A Planner, a Writer, and an Editor collaborate in sequence to create educational blog articles for students in Classes 6–10.

CLASSROOM STUDY NOTES

Project context: Ramanujan Classes
Example topic: Importance of Mathematics in Everyday Life

Inside these notes

- *Agentic AI and multi-agent systems*
- *Architecture and end-to-end workflow*
- *Planner, Writer, and Editor agents*
- *Agent, Task, Crew, and Process.sequential*
- *Task hand-off, API keys, and token usage*
- *Version 1.0 scope and interview questions*

1. UNDERSTANDING AGENTIC AI

Artificial Intelligence (AI) refers to computer systems that perform tasks such as understanding language, analysing information, or generating text.

An AI agent is an AI-powered worker configured with a goal and instructions. It receives information and produces an output or performs an assigned task.

Agentic AI describes systems that use agents to carry out goal-oriented steps, sometimes making decisions about how to complete their assigned work.

> *IN THIS PROJECT*
> *Three focused agents collaborate: Planner → Writer → Editor. CrewAI coordinates their work.*

2. WHAT IS A MULTI-AGENT SYSTEM?

A multi-agent system uses multiple agents, each with a defined responsibility, to work toward a shared outcome. Work is divided into smaller tasks instead of asking one agent to do everything.

- *Clear responsibilities: each agent has a focused role.*
- *Structured process: planning happens before writing; editing happens after writing.*
- *Task hand-off: later tasks use outputs from earlier tasks.*

> *IMPORTANT*
> *Your current workflow is a sequential three-agent content pipeline, not a fully autonomous publishing system.*

3. PROJECT ARCHITECTURE

USER INPUT	*Provides an educational topic*
↓	
CREWAI CREW	*Process.sequential*
↓	
PLANNER AGENT	*Creates article outline and key points*
↓ plan output	
WRITER AGENT	*Writes the complete article using the plan*
↓ write output	
EDITOR AGENT	*Reviews and refines the article*
↓	

FINAL MARKDOWN ARTICLE	*Human review → manual publishing*

Read the diagram from top to bottom. The user supplies a topic; CrewAI runs the tasks in order; a person reviews the final article.

4. THE THREE AGENTS

An agent is configured with a role, goal, backstory, and settings. These guide how the language model approaches its assigned work.

A. PLANNER AGENT — THE ORGANISER

- *Role: Educational Content Planner.*
- *Goal: Plan a clear article for students in Classes 6–10.*
- *Backstory: Contextualises the agent as an educational planner familiar with school topics.*
- *Output: Article structure, headings, key points, examples, and flow.*

B. WRITER AGENT — THE DRAFTER

- *Role: Educational Content Writer.*
- *Goal: Turn the outline into a complete, student-friendly article.*
- *Backstory: Guides the agent to explain concepts in simple English with examples.*
- *Input: Planner task output. Output: Complete first draft.*

C. EDITOR AGENT — THE REVIEWER

- *Role: Educational Content Editor.*
- *Goal: Improve clarity, grammar, structure, and suitability for the audience.*
- *Backstory: Guides the agent to preserve meaning while improving quality.*
- *Input: Writer task output. Output: Polished article.*

> *ROLE vs GOAL vs BACKSTORY*
> *Role = who the agent acts as. Goal = what it should achieve. Backstory = context that shapes its approach.*

5. CORE CREWAI BUILDING BLOCKS

Agent: Defines an AI worker with a role, goal, backstory, and settings.

Task: Describes a piece of work, its expected output, and the agent responsible.

Crew: Groups agents and tasks and coordinates their execution.

Process.sequential: Runs tasks one after another in the order supplied.

kickoff_async(): Starts the Crew asynchronously. In Colab, you used await crew.kickoff_async(inputs={...}) because the notebook environment already runs an event loop.

6. END-TO-END WORKFLOW

STEP 1 — USER INPUT

You provide a topic, for example: Importance of Mathematics in Everyday Life. It is passed in a dictionary using the key topic.

STEP 2 — PLANNER TASK

The Planner prepares an outline. Its output becomes context for the Writer.

STEP 3 — WRITER TASK

The Writer uses the topic and plan to produce the full article with sections, examples, and explanations.

STEP 4 — EDITOR TASK

The Editor receives the draft and improves grammar, clarity, structure, and audience suitability.

STEP 5 — FINAL RESULT

CrewAI returns the final task result. You inspect it and decide whether it is ready to publish.

> *KEY IDEA: SEQUENTIAL EXECUTION*
> *Planner → Writer → Editor. The Writer needs the plan, and the Editor needs the draft.*

7. MAIN CODE PATTERNS

Agent definition

Agent(role=..., goal=..., backstory=..., allow_delegation=False, verbose=True)

Creates a configured agent. allow_delegation=False disables agent delegation; verbose=True enables detailed execution logs.

Task definition

Task(description=..., expected_output=..., agent=planner)

Assigns a defined piece of work to an agent. The description explains the work; expected_output describes the desired result.

Crew definition

Crew(agents=[planner, writer, editor], tasks=[plan, write, edit], process=Process.sequential, verbose=True)

Groups agents and tasks and specifies sequential execution.

Run the workflow

result = await crew.kickoff_async(inputs={"topic": "Importance of Mathematics in Everyday Life"})

Starts the workflow with the supplied topic and returns the Crew result.

8. DATA FLOW BETWEEN TASKS

- *Topic is the initial input.*
- *Plan output gives the Writer its structure.*
- *Draft output gives the Editor the article to review.*
- *Edited output becomes the final result.*

> *REMEMBER*
> *The quality of later stages depends partly on the clarity and usefulness of earlier outputs.*

9. API KEY & TOKEN USAGE

The OpenAI API key lets CrewAI send requests to a language model.

UTILS.PY AND getpass()

Your utility function asks for the API key without displaying the typed characters in the notebook cell. The notebook imports the function and places the returned value into the OPENAI_API_KEY environment variable.

Example pattern:

```
from getpass import getpass
def get_openai_api_key():
    return getpass('Enter your OpenAI API Key:')
```

- *Restarting Colab alone does not use model tokens.*
- *Defining Python objects does not normally make a model request.*
- *Running the agents makes model requests and consumes tokens.*
- *Displaying an already-generated result does not make another model request.*
- *Running the Crew again makes new requests and uses additional tokens.*

> *SECURITY TIP*
> *Never hardcode the real API key or commit it to GitHub. Clear notebook outputs and inspect the notebook before uploading.*

10. VERSION 1.0 — SCOPE & BOUNDARIES

Implemented: topic input; Planner → Writer → Editor sequence; Markdown article generation; manual human review.

Not yet implemented: live web research, external source citations, automated fact-checking, and automatic website publishing.

> *ACCURATE PROJECT DESCRIPTION*
> *Describe this as a three-agent educational content generation workflow. Do not claim live web research or automatic publishing until those features are built.*

11. YOUR EXAMPLE OUTPUT

Your example topic was Importance of Mathematics in Everyday Life. The article covered managing money, fractions and cooking, time and travel, geometry, algebra, sports and data, technology, student practice questions, and advice for parents.

You manually reviewed the generated article before considering it ready for publication.

12. QUICK REVISION SHEET

- *Agent: A configured AI worker with a role and goal.*
- *Task: A defined unit of work assigned to an agent.*
- *Crew: A group of agents and tasks coordinated together.*
- *Sequential process: Tasks execute one after another.*
- *LLM: Large Language Model; generates language-based responses.*
- *Token: A unit of text processed by a language model.*

13. INTERVIEW QUESTIONS & ANSWERS

1. What is Agentic AI?

AI agents work toward goals by carrying out assigned steps or tasks.

2. Why use multiple agents?

To divide planning, writing, and editing into focused responsibilities.

3. What is CrewAI?

A framework for orchestrating role-based AI agents and tasks.

4. What does the Planner do?

Creates the outline and key points that guide article writing.

5. What does the Writer do?

Uses the plan to generate the complete educational article.

6. What does the Editor do?

Improves grammar, clarity, structure, and readability.

7. What is Process.sequential?

It executes the Crew's tasks in the order they are provided.

8. Agent vs Task?

An Agent is the worker; a Task is the work assigned to that worker.

9. Why use getpass()?

It hides typed characters when collecting the API key.

10. Does this version perform live web research?

No. The current three-agent setup does not have a live web-search tool configured.

11. Does it publish articles automatically?

No. A person reviews and publishes the generated article manually.

12. What happens if the Crew runs again?

It makes new model requests, which consume additional tokens.

PROJECT SUMMARY

I built a CrewAI-based multi-agent workflow in Python. The Planner creates an outline, the Writer generates the article, and the Editor refines it. The Crew executes these tasks sequentially and returns a Markdown article for human review.

END OF VERSION 1.0 NOTES